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1. Introduction

In this project, I have created and implemented LAN and WAN for Company A on Cisco Packet Tracer. For Company A, there were separate departments like Engineering, Manufacturing, Sales Marketing, Administration, Internal Servers, DMZ, and Guest WLAN. After creating the LAN, LAN from Company A was further connected with the LAN of Company B and Company C using WAN.

The objective of this project was to develop an entire network design, which would include VLSM Subnetting, VLAN, Inter VLAN routing, DMZ, guest WLAN, and WAN Routing. However, efficiency was another major component of this project as the router available only three Ethernet interfaces.

Company A has 230 employees divided into four departments:

- Engineering – 100 employees
- Manufacturing – 80 employees
- Sales/Marketing – 30 employees
- Administration – 20 employees

The LAN is also made up of a DHCP server, a DNS server, FTP server 1, FTP server 2, Web server, and Email server. Web and Email servers are located in the DMZ, and there is also a guest WLAN available for visitors.

2. Company A Subnet Allocation

Company A has been allocated the IP addresses from 216.244.168.0 to 216.244.175.255. VLSM was used in subdividing the network according to the number of devices required per subnet. It is the most efficient approach since each department varies in size, so using VLSM avoids wasting IP addresses.

Subnet	Network ID	Subnet Mask	CID R	Broadcast	First Host	Last Host	Max Hos ts	Default Gateway
Engineering	216.244.168.0	255.255.255.128	/25	216.244.168.127	216.244.168.1	216.244.168.126	126	216.244.168.1
Manufacturing	216.244.168.1	255.255.255.1	/25	216.244.168.	216.244.168.1	216.244.168.	126	216.244.168.1

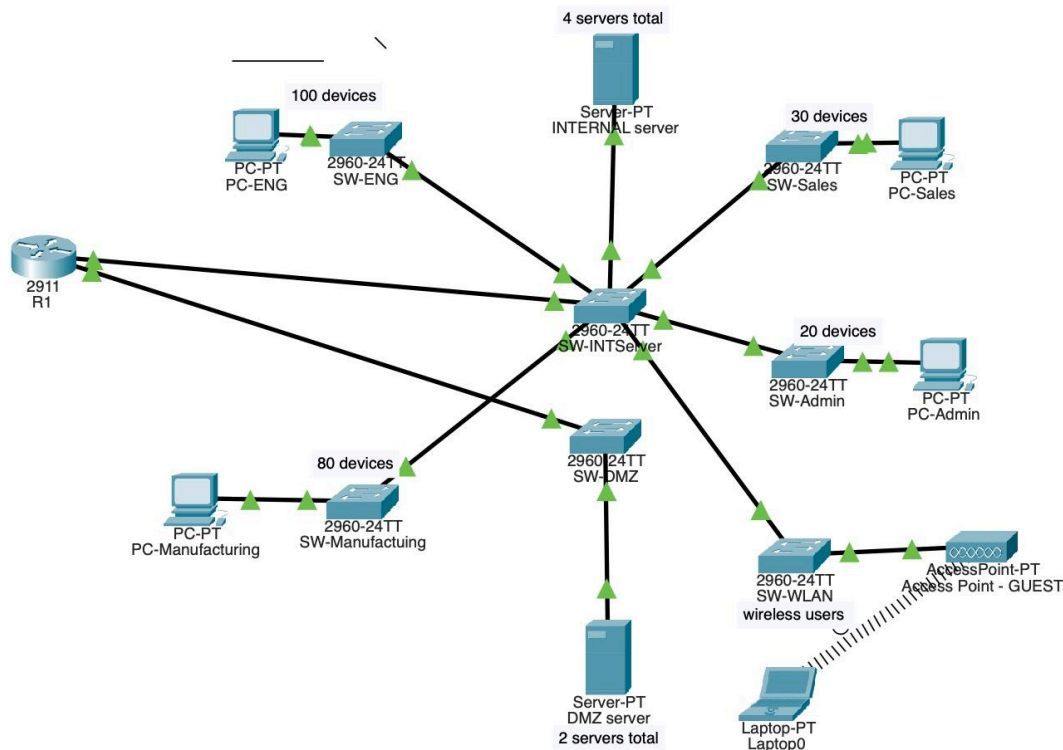
	28	28		255	29	254		29
Sales	216.244.169.0	255.255.255.192	/26	216.244.169.63	216.244.169.1	216.244.169.62	62	216.244.169.1
Administration	216.244.169.64	255.255.255.24	/27	216.244.169.95	216.244.169.65	216.244.169.94	30	216.244.169.65
Internal Server	216.244.169.96	255.255.255.240	/28	216.244.169.111	216.244.169.97	216.244.169.110	14	216.244.169.97
DMZ	216.244.169.112	255.255.255.248	/29	216.244.169.119	216.244.169.113	216.244.169.118	6	216.244.169.113
Guest WLAN	216.244.169.128	255.255.255.192	/26	216.244.169.191	216.244.169.129	216.244.169.190	62	216.244.169.129
WAN	216.244.175.252	255.255.255.252	/30	216.244.175.255	216.244.175.253	216.244.175.254	2	216.244.175.253

3. Cisco Packet Tracer Topology

In Cisco Packet Tracer, the LAN for Company A was created by using one router, several switches, one switch for the internal side, one switch for the DMZ, and one access point for the wireless side. Only one device was used to represent a subnet, with a note showing the total number of devices on a subnet.

The representative devices connected are:

- 1 PC for Engineering
- 1 PC for Manufacturing
- 1 PC for Sales/Marketing
- 1 PC for Administration
- 1 Internal Server
- 1 DMZ Server
- 1 wireless laptop for the Guest WLAN



4. VLAN, DMZ, Guest WLAN, and WAN Design

VLANs were used to separate the internal company networks. This was necessary because Company A has multiple subnets, but the router only has three Ethernet ports. Instead of using a separate physical router interface for every subnet, one router interface was used as a trunk link, and router subinterfaces were configured for the different VLANs.

The VLANs were used to segregate the internal networks within the company. This was essential because Company A had several subnets. However, the router only had three Ethernet ports, so rather than assigning individual router interfaces to every subnet, a single router interface was used as the trunk link and subinterfaces were assigned to different VLANs.

The VLANs used in the design are:

VLAN	
10	Engineering
20	Manufacturing

30	Sales/Marketing
40	Administration
50	Internal Server
70	Guest WLAN

This design was optimal because it saves router ports, keeps departments separated, and still allows communication between VLANs through the router.

The DMZ was used for the Web server and Email server. This was a better design than placing those servers directly inside the internal LAN because public-facing services should be separated from internal company devices.

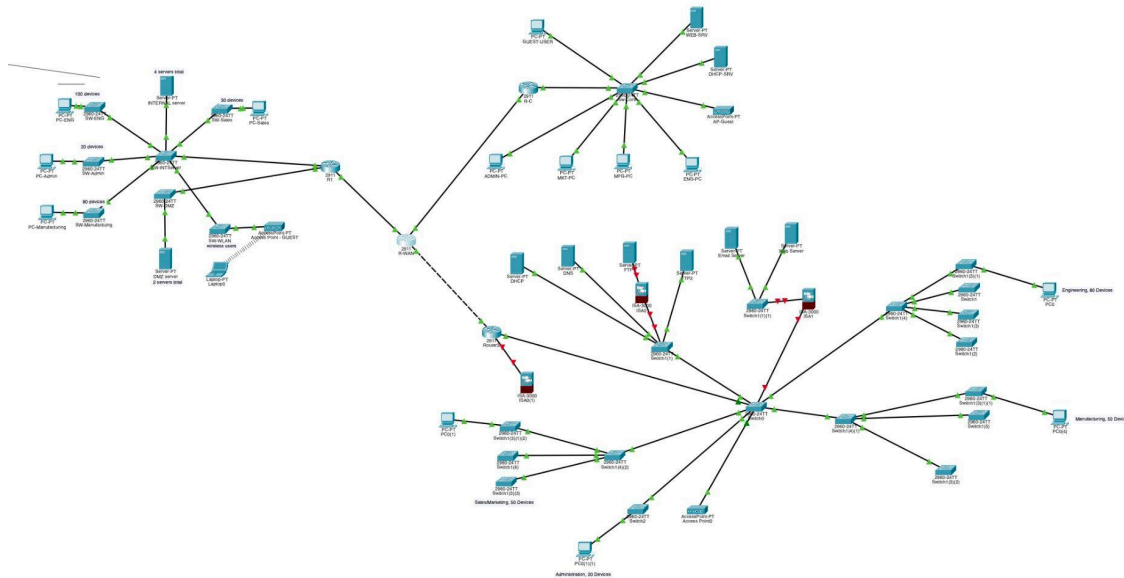
The Guest WLAN was placed in its own subnet. This keeps visitor wireless users separate from employee devices and internal servers.

```
Router>show ip interface brief
Interface                IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0       216.244.169.113 YES manual up                  up
GigabitEthernet0/1       unassigned      YES manual up                  up
GigabitEthernet0/1.10    216.244.168.1   YES manual up                  up
GigabitEthernet0/1.20    216.244.168.129 YES manual up                  up
GigabitEthernet0/1.30    216.244.169.1   YES manual up                  up
GigabitEthernet0/1.40    216.244.169.65  YES manual up                  up
GigabitEthernet0/1.50    216.244.169.97  YES manual up                  up
GigabitEthernet0/1.70    216.244.169.129 YES manual up                  up
GigabitEthernet0/2       216.244.175.254 YES manual up                  up
Vlan1                    unassigned      YES unset  administratively down down
```

For the WAN, I used a central router called R-WAN. Company A, Company B, and Company C connect to this central WAN router. This design was simple and organized because each company router only needed one WAN connection.

The WAN connections are:

WAN Link	Company Router IP	R-WAN IP
Company A to R-WAN	216.244.175.254	216.244.175.253
Company B to R-WAN	216.244.183.254	216.244.183.253
Company C to R-WAN	216.244.254.254	216.244.254.253



```

Router>show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       216.244.175.253 YES manual  up          up
GigabitEthernet0/1       216.244.254.253 YES manual  up          up
GigabitEthernet0/2       216.244.183.253 YES manual  up          up
Vlan1                    unassigned      YES unset   administratively down down

```

5. Routing Method

For the WAN routing, we considered static routing, RIPv2, and OSPFv2.

Static Routing:

Advantages

- simple
- easy to control
- good for a small network

Disadvantages

- routes must be added manually
- not good for a large network

RIPv2:

Advantages

- automatically shares routes
- easier than OSPF

Disadvantages

Slower

not as efficient for a bigger WAN

OSPFv2:

Advantages

faster

better for larger networks

Disadvantages

more complicated to configure

Selected Method

For this project, we used static routing. we chose it because the WAN is small and it is easier to manage with manual routes.

6. Results

7.1 Company A to Company C Result

In order to confirm WAN connectivity for Company A, I performed a ping test from the Company A router to the Company C WAN IP address. Ping was successful, proving that Company A had access to Company C through the WAN network.

```
Router#ping 216.244.254.254

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 216.244.254.254, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

7.2 Company B WAN Result

To test the direct WAN connection to Company B, I did the ping test from R-WAN to the Company B router and it was successful

```
Router>ping 216.244.183.254

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 216.244.183.254, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms
```

7.3 Company B to Company A Result

In order to confirm the communication between Company B and Company A, I performed pings on both the Company B router and the WAN IP and LAN Gateway of Company A. The results were successful.

```
Router>ping 216.244.175.254
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 216.244.175.254, timeout is 2 seconds:
```

```
!!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```

```
Router>ping 216.244.168.1
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 216.244.168.1, timeout is 2 seconds:
```

```
!!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

7.4 Company B to Company C Result

In order to confirm that there was communication between Company B and Company C, pings were done from the Company B router to the WAN IP of Company C and the LAN Gateway of Company C.

```
Router>ping 216.244.254.254
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 216.244.254.254, timeout is 2 seconds:
```

```
!!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

```
Router>ping 216.244.184.1
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 216.244.184.1, timeout is 2 seconds:
```

```
!!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

The results showed that:

- R-WAN has routes for all three companies,
- Company B router can reach Company A and Company C,
- Company C router can reach Company A and Company B.

This shows that the WAN is working correctly.

7. Network Concepts and design

8.1 Switches, Hubs, Routers, Access Points

A switch is a Layer 2 device that forwards frames using MAC addresses. It learns which MAC is on which port and only sends a frame to the right port. Each port is its own collision domain. This is why every department in this design uses a switch.

A hub is a Layer 1 device that just repeats every signal to all its other ports. All ports share one collision domain so collisions are common. No hubs are used here for that reason.

A router is a Layer 3 device that forwards packets between different IP networks using its routing table. It also separates broadcast domains. Each company has one router, plus the central R-WAN.

An access point (AP) lets wireless devices connect to the wired LAN. The guest WLAN uses one AP, which can support up to 64 wireless users.

8.2 Collision domains, broadcast domains

A collision domain is the area where two devices can transmit at the same time and cause a collision. Every switch port is its own collision domain, while a hub puts everything into one. Using switches everywhere in the LAN basically eliminates collisions.

A broadcast domain is the area where a broadcast frame (like ARP) reaches every device. Switches alone do not separate broadcasts. Routers do, and so do VLANs. In this design, VLANs put each department into its own broadcast domain even though they share the same physical switches, which keeps each department's broadcast traffic away from the others.

8.3 Static routing, dynamic routing, and selective routing algorithms

Static routing means the administrator types the routes in by hand. It is predictable, uses no extra bandwidth, and is best for small fixed networks. It does not scale.

Dynamic routing means routers exchange information automatically using a protocol like RIPv2 or OSPFv2. The routers learn the topology on their own and react to changes (a link going down, for example), but they use more CPU and bandwidth and are harder to configure.

Selective routing (or policy-based routing) picks the path based on more than just the destination IP — for example the source address or the type of traffic. It is useful when certain traffic needs to take a special path.

8. Detailed Traffic Flow

(a) An employee in the engineering department pings the FTP server 1 of his/her own company

For this project since only one representative internal server is used for the internal server subnet. That server represents DHCP, DNS, FTP Server 1, and FTP Server 2.

IP Address of the Engineering PC is 216.244.168.2, and the gateway address is 216.244.168.1. IP Address of the internal server is 216.244.169.98, and its gateway is 216.244.169.97.

Since the two devices are on different subnets, the PC will send the packet to its default gateway. To get the MAC address of the gateway, the PC uses ARP, and then sends the ICMP echo request to the router.

The packet will travel from the Engineering PC to Engineering switch, from there it will travel to the main internal switch and then to the router where the routing will take place from Engineering VLAN to Internal Server VLAN. Now, the packet will move back through the main internal switch and will reach the internal server which will send the response packet ICMP Echo Reply.

Path:

Engineering PC → SW-ENG → SW-INTServer → R1 → SW-INTServer → Internal Server

Protocols used:

ARP, ICMP, IP, Ethernet

(b) An employee in the engineering department pings the FTP server 2 of his/her own company/university.

This process is same as part (a), because FTP Server 2 is also in the internal server subnet.

FTP Server 1 is meant for Engineering and FTP Server 2 is meant for other departments, but for a basic ping, the path is the same unless access rules are added.

Path:

Engineering PC → SW-ENG → SW-INTServer → R1 → SW-INTServer → Internal Server subnet

Protocols used:

ARP, ICMP, IP, Ethernet

(c) Administration employee pings the web server of his or her own company

IP address of the Administration PC is 216.244.169.66 and the gateway is 216.244.169.65. Web server can be represented by the DMZ server whose IP address is 216.244.169.114.

Since the two devices are on different subnets, the PC will send the packet to its default gateway. To get the MAC address of the gateway, the PC uses ARP, and then sends the ICMP echo request to the router.

The router forwards the packet from the Administration VLAN to the DMZ subnet. The packet goes to the DMZ switch and then to the web server. The web server sends the ICMP Echo Reply back through the router.

Path:

Administration PC → SW-ADMIN → SW-INTServer → R1 → SW-DMZ → DMZ Server/Web Server

Protocols used:

ARP, ICMP, IP, Ethernet

(d) An employee in the administration department pings the web server of another company.

In this case, the Administration PC is trying to reach a server in company C, so the traffic must go through the WAN.

The Administration PC first sends the packet to its default gateway 216.244.169.65. Again Arp is used first if the MAC address is not already known. Then the ICMP Echo Request goes to the Company A router.

The Company A router checks its routing table and sends the packet to the central WAN router, R-WAN. R-WAN then forwards it to the Company C router, Then company C router forwards it to the correct server subnet. The reply comes back through the same path.

Path:

Administration PC → SW-ADMIN → SW-INTServer → R1 → R-WAN → other company router → other company web server

Protocols used:

ARP, ICMP, IP, Ethernet, and static routing

9. VPN

A VPN is a secure communication channel between two computer networks (remote user and network). It functions by encapsulating each IP packet into another one and encryption it before transmission. Then, both parties must establish identity using pre-shared keys or certificates, generate encryption keys, and then exchange actual data through the protected channel.

The two most popular types of this kind of protection are IPsec and SSL. IPsec operates on the network layer and it is typically used to secure communication between routers (site-to-site VPN). In turn, SSL operates on a higher layer and it is commonly used when only a single remote user connects to the network through a client or browser.

In order to provide security, a tunnel must have three main properties. Firstly, there is confidentiality because all of the information is sent using encryption, hence the attacker can only capture the ciphertext. Secondly, packets are hashed so that the attacker cannot modify them, so any modifications will cause the packet to be discarded. Thirdly, authentication proves identities of the sender and receiver, which prevents attackers from impersonation.

10. Conclusion

Company A was successfully built in Cisco Packet Tracer with a VLSM subnet plan, VLANs and router subinterfaces for each department, an internal server subnet, a DMZ for the web and email servers, and a guest WLAN. The three companies were then connected through a central R-WAN router using static routing, which was the right choice for a small fixed network. Ping tests across the WAN confirmed that any host in one company can reach any host in another. With a VPN added on top, remote users can also connect securely to the LAN over the public Internet.